

# The *InGrigna!* project

**Marco Corvi**

*InGrigna!.*

*Via Borgo Antico n ° 7 Recco GE 16036 (Italia) marco.corvi@gmail.com"*

## **Abstract**

The InGrigna! caving project started in 2002 when caving clubs of Lombardy, Italy, joined their efforts to pursue the exploration of a number of newly discovered abysses by the Speleo Club Erba in the Grigna mountain. At present nine caving clubs participate to InGrigna! as well as individual speleologists, for a total of about 60 people. The project's objectives are the search of new caves, exploration and surveying of the caves, documentation (papers, photos, videos) of the speleological activity, and collection of scientific data (biospeleology and paleontology). InGrigna! operates mostly in the areas of Tivano and Grigna (Northern Italy). In the former the cave system of Valle del Nose, which is over 62 Km in length, has been explored. In Grigna the number of known caves has been doubled in the past twelve years, to over 1100, as well as the number of known deep caves. The most important achievement is the identification, through the discovery of several junctions between the caves, of the cave system of Releccio, joining 14 caves with 18 entrances, and over 22 km long. The latest news is the recent discovery of the continuation at the bottom of W le Donne, the deepest cave of the system, which is now -1260 m with open exploration leads.

**Keywords.** Grigna, Tivano, W le Donne, Releccio, Valle del Nose.

# El projecte *InGrigna!*

## **Resum**

El InGrigna! projecte d'espeleologia va començar el 2002, quan els clubs d'espeleologia de Llombardia, Itàlia, van unir els seus esforços per prosseguir l'exploració d'una sèrie d'avencs acabades de descobrir pel Espeleo Club de Erba, a la muntanya Grigna. Actualment nou clubs d'espeleologia participen a InGrigna! així com espeleòlegs individuals, per a un total de prop de 60 persones. Els objectius del projecte són la recerca de noves coves, l'exploració i prospecció de les coves, documentació (documents, fotos, vídeos) de l'activitat espeleològica, i la recollida de dades científiques (bioespeleologia i paleontologia). InGrigna! opera principalment en les àrees de Tivano i Grigna (nord d'Itàlia). En el primer cas el sistema de coves de la Vall del Nose, que és de més de 62 km de longitud, ha estat explorat. En Grigna el nombre de coves conegudes s'ha duplicat en els últims dotze anys, a més de 1.100, així com el nombre de coves profundes. L'assoliment més important és la identificació, a través del descobriment de diversos encreuaments entre les coves, del sistema de coves de Releccio, unint-se a 14 coves amb 18 entrades, i més de 22 km de llarg. L'última notícia és el recent descobriment de la continuació a la part inferior de W li Donne, la cova més profunda del sistema, que és ara 1260 m amb vies d'exploració obertes.

**Paraules clau.** Grigna, Tivano, W le Donne, Releccio, Valle del Nose.

# El proyecto InGrigna!

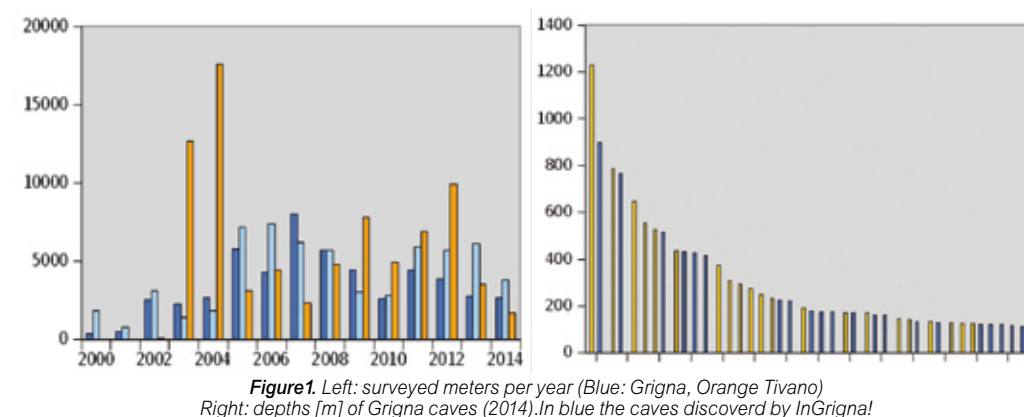
## Resumen

El proyecto InGrigna! de espeleología comenzó en 2002, cuando los clubes de espeleología de Lombardía, Italia, unieron sus esfuerzos para proseguir la exploración de una serie de simas recién descubiertas por el Espeleo Club de Erba, en la montaña Grigna. Actualmente nueve clubes de espeleología participan en InGrigna! así como espeleólogos individuales, para un total de cerca de 60 personas. Los objetivos del proyecto son la búsqueda de nuevas cuevas, la exploración y prospección de las cuevas, documentación (documentos, fotos, vídeos) de la actividad espeleológica, y la recogida de datos científicos (bioespeleología y paleontología). InGrigna! opera principalmente en las áreas de Tivano y Grigna (norte de Italia). En el primer caso el sistema de cuevas del Valle del Nose, que tiene más de 62 km de longitud explorados. En Grigna el número de cuevas conocidas se ha duplicado en los últimos doce años, a más de 1.100, así como el número de cuevas profundas. El logro más importante es la identificación, a través del descubrimiento de varios cruces entre las cuevas, del sistema de cuevas de Releccio, uniéndose 14 cuevas con 18 entradas, y más de 22 km de recorrido. La última noticia es el reciente descubrimiento de la continuación en la parte inferior de W le Donne, la cueva más profunda del sistema, que tiene ahora 1260 m con vías de exploración abiertas.

**Palabras clave.** Grigna, Tivano, W le Donne, Releccio, Valle del Nose.

## The InGrigna! project

The speleological research in the Grigna massif (northern Italy) began in the second half of the past century. The first important discoveries came in the '80s with the exploration of the Abisso Maron Glaces, -557m. After the exploration of W le Donne to a depth of 1180 m, in the early '90, the speleological activity declined. Probably this was due to the caving club based approach to speleology, principally in search of the record depth, competing one another. In 2000 the Speleo Club Erba started to investigate the area with a new frame of mind: a methodical approach that lead them to look for caves in places that have been ignored for decades. And they found several deep caves. Conscious that the club did not have enough resources to carry forward the explorations, they proposed to other clubs to join the effort. Wherefore the caving project InGrigna! was founded in 2002. From the very beginning the project has been characterized by its openness to participation, a systematic approach to the exploration, and sharing of entire data, results, and informations among the participants. The project is open to anyone willing to contribute, help the explorations, and share the results. At the moment there are nine caving clubs and some independent speleologists comprising about 60 people actively participating. No registration is required, and there is no fee. The budget of the project relies on donations, spontaneous contributions by the participants, and incomes from gastronomical stands held at annual national conventions. The InGrigna! project is active mostly in the areas of Tivano and Grigna, and takes part in several projects of the regional speleological federation (FSLi). The approach to speleology is methodical. The whole area of Tivano and Grigna are thoroughly searched for entrances, even on the steep mountain sides. The caves are systematically revisited. All the possible leads are checked. Every passage is surveyed. The cave surveys are collected in a Compass project. The data are organized in a GIS. Finally thematic maps of the area are regularly prepared for the work on the field. The speleological activity is documented in articles, photos, and videos. Most of the articles are written for caving club bulletins. La Grigna al Contrario is compiled by the project specifically concerning the explorations in Grigna. Articles and short notes are also regularly published in Speleologia, the magazine of the Italian Speleological Society. Photos and videos are used for presentations at caving meetings and for the general public. At the end of each year the project assembles all the compiled material about Grigna for the Parco delle Grigne, the institution that administers the Grigna area. In Tivano new caves have been discovered (eg, Terzo Mondo and Buco del Latte). However the most important results came from the exploration of Ingresso Fornitori and Stoppani. Three conjunction passages between Fornitori and Stoppani, and one between Stoppani and Tacchi-Zelbio were found. These discoveries lead us to confirm the existence of the Valle del Nose cave system which currently comprises 62.4 km of surveyed passages.



The results of the project in Grigna mountain are no less relevant. The number of known caves has been increased from about 600 to over 1100. Clearly most of them are small caves. But also several deep new caves or continuations have been found (Kinder Brioschi e i Cinque Minerali -900 m, Abisso delle Spade -767 m, Voragine di Oltre 40 m -514 m), as shown with the blue bars in Figure 1 on the right. The meters of survey per year for the caves in Grigna and in Tivano are shown on the left in Figure 1. The light blue bars are the number of new caves found in Grigna (scaled by 100). The most important achievement in Grigna is the identification of the cave system of Releccio, which comprises 14 caves with 18 entrances, obtained through the discovery of junctions between the caves, two of them at a depth of -900 m (Kinder-W le Donne, Orione-W le Donne). This was made possible thanks to the accurate survey work which provided the clues for the places to pursue. The cave system is now 22.8 km long and -1260 m deep.

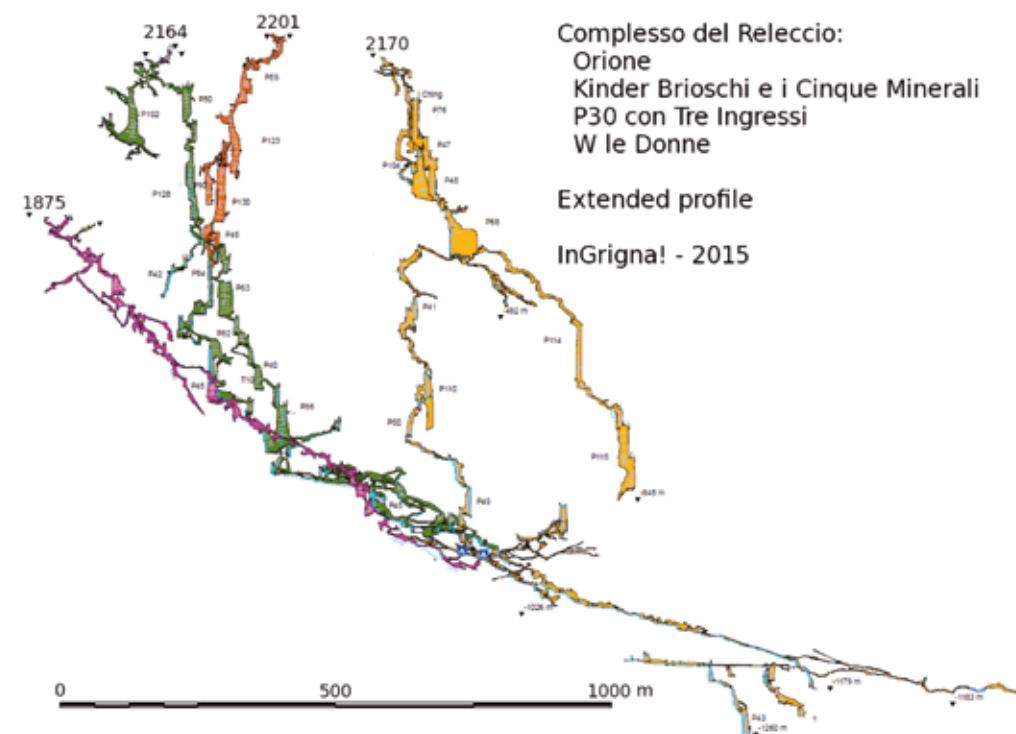


Figure 2. Releccio cave system - Abridged extended profile. (A. Maconi)

Very recently a continuation has been found at the bottom of W le Donne which opens at a height of 2170 m asl. The geological structure of the Grigna is dominated by a syncline and the cave W le Donne is known to have a hydrological connection with the spring of Fiumelatte at a distance of about 8 km from the bottom of the cave and 700 m below. The work in W le Donne started in 2008 with the completely new rigging on the pits. The old equipments left by the previous explorers have all been taken out of the cave. Since then the search for the continuation has begun with a thorough revision of the galleries at -1100 m. The sump at -1150 m has been passed by D. Corengia, first in Dec. 2011, and a second time in Nov. 2012, when he stopped at the top of a 20m deep pit. Two years later, Nov. 2014, a new way has been found after a series of tight passages leading to a new underground region, the exploration of which is under way. It seems that we have now reached the other side of the underground syncline. The cave is now -1230m deep (the system -1260m) and promises to continue.



*Cresta di Piancaformia. Along the way to W le Donne.  
(A. Maconi)*



*A wet passage at -1100 m in W le Donne.  
(D. Corengia)*

## References

Premazzi, A. et al. Tra i due rami del Lago di Como  
Speleologia 55 2006 14-27

aa. vv. Decenni di esplorazioni... L'11 febbraio 2012 si è aperta la porta magica!  
Speleologia 66 2012 11-17

Premazzi, A. et al. Il complesso del Releccio. Cronaca e storia dell'evoluzione. Speleologia 62 2010 30-39

Corengia, D. Immersione nel sifone di -1150 m nell'Abisso di W le Donne (Grigna Lc)  
Speleologia 66 2012 74